# An energy budget

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Connect work, motion, heating and an honest model conclusion.

**SUCCESS CRITERIA - I can...**

1. Keep all terms in consistent energy units.
2. Distinguish mechanical-energy change from total-energy loss.
3. Report assumptions with the energy budget.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

A 0.3 kg object moving at 5 m/s stops through friction over 0.7 m. Find energy transformed and average opposing-force magnitude.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $E_i + E_{in} = E_f + E_{out}$.
i/f = initial/final stored energy; in/out = transfers across the chosen system boundary. Account for every form.

**Worked reasoning:** An input of 100 J gives 60 J useful motion, 25 J thermal transfer and 15 J sound/other transfers. The account balances because 60+25+15=100 J.

**Your turn:** Draw proportional energy arrows and calculate the useful fraction.

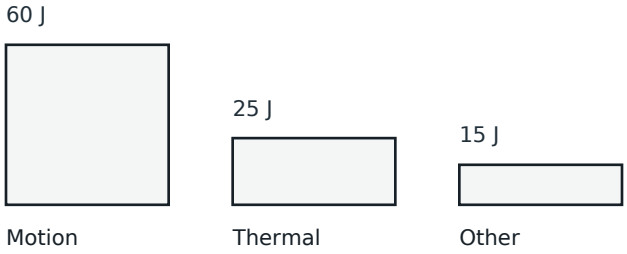
**Initial** **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** ① ② ③

## 03 Read and interpret the model

A modeled system starts with 100 J. It ends with 60 J motion energy, 25 J thermal energy and 15 J sound/deformation. Draw an energy account and check the total.




Bar height represents energy. Read the value and unit above each bar.

**Marzano check - circle:** ① ② ③

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Continue down each column.

## 04 Apply the physics

A simulated 1000 kg probe has K = 8 × $10^6$ J and gravitational U = -6 × $10^6$ J, with U = 0 at infinity. Find total mechanical energy. Drag then removes 3 × $10^6$ J. Complete the energy account and discuss whether the final state is bound in an isolated central-gravity model.

**Initial** **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** ① ② ③

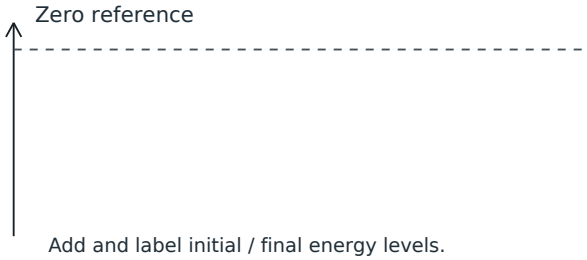
## 05 Explain your reasoning

What information is missing if an impact report gives only total kinetic energy and says all of it heats the ground?

**Marzano check - circle:** ① ② ③

## 06 Review: connect an earlier idea

With GMm=192 J·m, move an object from r=4 to r=8 m. Find work by gravity using U=-GMm/r.
Label the distances or energy levels used in your reasoning.




Add and label initial / final energy levels.

**Marzano check - circle:** ① ② ③

## 07 Exit check

A 2 kg cart descends 3 m from rest with g=10. Losses are 12 J. Find bottom speed. Use the energy model to show the quantities you compare; label units and any unknowns.

**Initial** **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** ① ② ③

## 08 STRETCH: test the model

A device takes 600 W for 5 s and raises a 10 kg load 12 m using g=10. Find useful work, efficiency and remaining energy. Use the energy model to show the quantities you compare; label units and any unknowns.

**Initial** **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** ① ② ③